

Study and Research Focus

Digital system design (RTL, FPGA and ASIC flows), embedded real-time systems, and data-driven systems. Applying for a master's programme in Electrical Engineering and related fields, with interest in laboratory research and assistant roles.

Education

2023–2027 **Sharif University of Technology, Tehran, Iran**

B.Sc., Electrical Engineering (Digital Systems focus) • expected Aug 2027

- **Cumulative GPA: 16.88/20.** Above the department average (16.49) and university average (16.09) for this cohort, as printed on the [Sharif transcript](#).
- **Class rank: 90 of 200 students** in the 2023 Electrical Engineering entry cohort (Sharif registrar certificate).
- Most recent semester (Spring 2026): **19.30/20** over 19 units.
- **National rank 134** among about 150,000 candidates in the Mathematics and Physics group of the Iranian University Entrance Exam (Konkur).
- **Relevant coursework (grade/20):** ASIC/FPGA System Design (19.4); Embedded Systems (19.8); Foundations of Data Science (19.3); Computer Interfacing, a graduate course (19.0); Logic Circuits and Digital Systems with lab (18.2); Microprocessor Systems Design (18.1).

Course Projects

Spring 2026 **Smart Surveillance System on an Embedded Linux Board** [source](#)

Embedded Systems course, Dr. Gholampour • Orange Pi Zero Plus 2 (H5), C, Python

- **System:** Individual project built around an on-board detector, an HTTPS dashboard, a REST API and MQTT/e-mail alerts.
- **Reliability and evaluation:** Added a watchdog and thermal throttling; measured boot time, detection accuracy and load.

Spring 2026 **OTP Memory Controller (RTL design, synthesis and verification)** [source](#)

ASIC/FPGA System Design course, Dr. Baharvand • Verilog, Cadence Genus and Xcelium

- **Design:** Co-designed a power-on FSM controller that reads a 256-bit OTP array, applies hardware patches and checks a CRC-16.
- **Team and flow:** Team of 4, led week 1 (rotating lead); verified with a testbench, then ran synthesis and gate-level simulation.

Spring 2025 **MIPS32 Processor Implementation on FPGA**

Computer Architecture lab, Dr. Movahedin • Verilog, MIPS32 assembly, Xilinx Zynq-7010

- **Design and validation:** Single-cycle, multi-cycle and pipelined MIPS32 CPUs; the multi-cycle one ran on an FPGA board.

2026–present **SUT Music: Data Platform and Recommendation Engine** [source](#)

Foundations of Data Science course, Dr. Salavati • Python, scikit-learn, PostgreSQL, FastAPI, React

- **Data:** Team of 3 (project lead); cleaned a Telegram music community's history into a database of 14,800+ tracks.
- **Recommenders:** Two generations (matrix factorization; two-stage ensemble) behind a backend API and a Telegram Web App.

Teaching Experience

- 2024–2026 **Teaching Assistant, Fundamentals of Programming and OOP**
Sharif University of Technology • Dr. Vosoughi Vahdat and Dr. Arasteh • four consecutive semesters
- Fall 2025 **Teaching Assistant, Electronics 1**
Sharif University of Technology • Dr. Fakharzadeh, Dr. Sarvari, Dr. Medi, Dr. Besharatirad
- Fall 2025 **Lab Assistant, Logic Circuit Lab (FPGA)**
Sharif University of Technology • Dr. Movahedin • supported students designing on Zynq-7010 FPGA boards
- Spring 2026 **Teaching Assistant, Theory of Circuits**
Iran University of Science and Technology (IUST)

Industrial Training

- Summer 2026 **Trainee, Parsian Zarin (ABLY): MQL5 expert advisors; Excellent Pass**

Software for Teaching and Community Service

- 2024–present **EE Department Bot Ecosystem and BotOS Framework**
Lead developer • Python (asyncio), PostgreSQL, Docker, Google Sheets API
- Began with the programming courses I assisted (Fundamentals of Programming, Object-Oriented Programming) and still maintain bots used by the EE department for course-file distribution, assignment grading, and re-judge ticketing, and by two student organisations; the platform has since been adopted by a second university (IUST).
 - Designed BotOS, a framework that deploys new bots from declarative JSON configuration in isolated Docker containers.

Technical Skills

- **Digital design:** Verilog, SystemVerilog, RTL design and testbenches; Xilinx Vivado, ModelSim/Questasim, Cadence Genus and Xcelium; Xilinx Zynq-7010.
- **Embedded and circuits:** C, assembly (MIPS32), embedded Linux on single-board computers; LTspice, Altium Designer.
- **Software and data:** Python (data analysis, machine learning, backend development), C++, Java, PostgreSQL, SQLite, Docker, Git, CI/CD, Linux.

Languages

Persian (native); English (official test score pending).